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Self-diffusion in a hyaluronic acid–albumin–water system as studied by NMR

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Hyaluronic acid (HA) is an anionic biopolymer that is present in many tissues and can be involved in cancerous neoformations. HA can form complexes with proteins (particularly, serum albumin) in the body. However, HA structures and processes involving HA have not been extensively studied by NMR because the molecule's rigid structure makes these studies problematic. In the current work, self-diffusion of HA and bovine serum albumin (BSA), and water in solutions was measured by ^{1}H pulsed field gradient NMR (PFG NMR) with a focus on the HA-BSA-D_2O systems at various concentrations of BSA and HA. It was shown that in the presence of even a small amount of HA, the self-diffusion coefficient (SDC) of BSA decreases. To explain this fact, three hypotheses were proposed and analyzed. The first one was based on the effect of slowing down of water mobility in the presence of HA. The second hypothesis suggested an effect of mechanical collisions of BSA with HA molecules. The third hypothesized that BSA and HA molecules form a complex where BSA molecules reduced in mobility. It was shown that the third mechanism is the most likely. The state of the BSA molecules in the BSA-HA-D_2O system corresponds to a 'fast exchange' condition from the NMR point of view: BSA molecules reside in the 'free' and 'bound' (with HA) states for much shorter time than the diffusion time of the PFG NMR experiment, 7 ms. The fractions of 'bound' BSA molecules in the BSA-HA complex were estimated.

Keywords: self-diffusion; nuclear magnetic resonance

Introduction

Hyaluronic acid (HA) is a copolymer of N-acetyl-D-glucosamine and D-glucuronic acid. HA is a high molecular weight, negatively charged polysaccharide found in almost all biological tissues. The molecule plays a diverse role in the body, having many structural, physiological and biological functions.[1,2] Its high capacity for lubrication, water sorption and retention results in its application in ophthalmic surgery.[3] Hydrogels based on HA have been considered as possible cell transplantation vehicles for regenerating diverse tissues. The formation of hydrogen bonds between HA chains may produce intermolecular (physical) associations that perturb both their structure and dynamics. For these reasons, HA is an attractive molecule for use in creating jelly-like structures, with possible applications in drug delivery and tissue engineering.[1]

Electrostatic repulsion between the carboxylate groups of HA favors an elongated configuration, with its persistence length reported to be in the range of 40–150 Å,[4] indicating that HA is a relatively rigid molecule.

Diffusion is one of the most fundamental properties of substances in solution. Therefore, there has been interest in understanding diffusion processes of biomacromolecules and water molecules in solutions and in hydrogels. In particular, HA diffusion has been measured by confocal fluorescence recovery after photobleaching and was found to decrease with increasing polymer concentration and to increase with increasing divalent ion concentration.[3] The self-diffusion coefficient (SDC) of HA, determined by dynamic light scattering measurements in a solution of 1 wt%, was significantly greater than that in solutions of neutral polymers at the same concentration.[2]

Diffusion also can be used as a tool in such studies because it allows the examination of the exchange of solutes between the gel and its surroundings, as well as within the crowded gel environment. Macromolecular solutes, such as enzymes in gel-forming systems, are of special interest.[5,6] HA can be successfully substituted for albumin as the sole macromolecule in human embryo transfer medium.[6] Electrophoretic measurements have shown that HA from synovial fluid, and bovine serum albumin (BSA) may form a complex at pH 8.6 in a veronal buffer, whereas decreased interaction occurs at pH 7.1–7.6 in a phosphate buffer.[7] Grampling *et al.* stated that the composition of the complex roughly fulfills the 'equivalence of charges' of HA and BSA.[8] They also showed that HA samples contain a fraction that could not complex with albumin.[8] Many details of this complex formation, including the dynamics of the molecules, remain unclear.

In this work, we applied pulsed field gradient (PFG) ^{1}H NMR to HA-water and HA-BSA-water systems to study the translation diffusion and intermolecular organization of these systems.

Materials and Methods

Materials

Hyaluronic acid sodium salt from rooster comb was used. The structure of the HA molecule is shown in Fig. 1. The weighted mean molecular mass of the HA is ~ $1.2 \cdot 10^6$ g/mol.[9]

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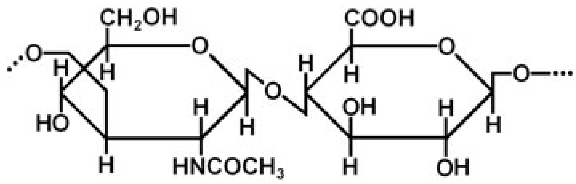

Figure 1. Structural formula of a monomeric unit of hyaluronic acid (HA).

Bovine serum albumin was purchased as a lyophilized powder. This peptide contains 607 amino acid residues and has a molecular mass of around 66 kDa and an isoelectric point of 4.7.

Hyaluronic acid, BSA and deuterated water (D_2O) were purchased from Sigma (St. Louis, MO).

Sample preparation

The compositions of the samples used are presented as percentages of HA and BSA in Table 1. BSA solutions were prepared by mechanically mixing powdered BSA with deuterated water. The solutions were transparent at all concentrations studied.

Hyaluronic acid solutions were prepared by placing the appropriate amount of solid HA sodium salt in a tube and loading the tube with the appropriate amount of D_2O. The process of swelling usually takes some time, so to attain complete homogenization, the samples were kept at room temperature for two to three days. After the swelling was complete, the samples were visually homogeneous, transparent and highly viscous.

Mixed HA–BSA solutions were prepared by combining the individually prepared BSA and HA solutions and allowing equilibration over two days. The pH of all studied samples was in the range 7.0–7.2.

Finally, 0.5-ml aliquots of the solutions were placed in 6-mm i.d. NMR tubes and sealed to prevent evaporation. Measurements were performed within one week of sample preparation.

Pulsed field gradient NMR

Experiments were performed using a homemade NMR device operating at 60 MHz ^{1}H NMR (1.4 T).[10] Diffusion decays were obtained as dependences of the spin-echo amplitude on the pulsed magnetic field amplitude, which varied from 0 to the maximum value of 30 T/m. For the stimulated echo pulse sequence used, the diffusion decay (DD) of the echo amplitude A can be described by the equation: [11]

$$A(2\tau, \tau_1, g, \delta) = \frac{I}{2}\exp\left(-\frac{2\tau}{T_2} - \frac{\tau_1}{T_1}\right)\exp\left(-\gamma^2\delta^2 g^2 D t_d\right), \quad (1)$$

where I is the factor proportional to the proton amount in the system; T_1 and T_2 are spin–lattice and spin–spin relaxation times, respectively; τ and τ_1 are periods in the pulse sequence; γ is the gyromagnetic ratio for protons; g and δ are amplitude and duration of the gradient pulse, respectively; $t_d = (\Delta\text{-}\delta/3)$ is the diffusion time; $\Delta = (\tau + \tau_1)$ is the time delay between the two gradient pulses; and *SDC* is the self-diffusion coefficient of the molecules.

Table 1. Compositions of hyaluronic acid (HA), bovine serum albumin (BSA) and HA + BSA solutions in D_2O (wt%)

Sample N	% BSA	% HA
1	4	—
2	2	—
3	1	—
4	—	2
5	—	4
6	—	8
7	4	0.25
8	4	0.75
9	2	0.25
10	2	0.75
11	1	0.25
12	1	0.75

Experimental Results

To understand the phenomena occurring in the system BSA–HA–D_2O, the corresponding systems prepared from two components, HA–D_2O and BSA–D_2O were studied first.

Diffusion in a HA–D_2O system

Figure 2 shows DDs in a HA–D_2O system at 4 wt% HA obtained with a diffusion time of 7 ms, $\delta = 0.16$ ms and at two different temperatures. Both curves demonstrate two-component behavior typical for macromolecular solutions.[12] The initial fast-decaying part is caused by the appearance of water protons in D_2O after partial deuterium substitution with protons from HA, whereas the slow-decaying part dominating at values of the abscissa larger than $4{\cdot}10^{-9}$ s/m^2 is caused by HA diffusion. The diffusion coefficients obtained from two-component analysis are shown in Table 2. Water diffusion coefficients were estimated with high precision. The SDCs of water are $1.55{\cdot}10^{-9}$ m^2/s and $2.3{\cdot}10^{-9}$ m^2/s at 30 °C and 45 °C, respectively. They decreased by a factor of 1.6 in comparison with bulk water SDCs.

Decays for the part of the DDs because of HA diffusion changed in amplitude only very slightly, so a pulsed field gradient was applied for as long as possible, 1 ms (Fig. 2, insert). Even under these conditions, the decay of the echo amplitude was small. Therefore, the value obtained for SDC gives only a rough estimation of the mean SDC value for HA as $\leq 10^{-13}$ m^2/s.

The apparent fraction of the DD components for HA increases with rising temperature because ^{1}H T_2 and T_1 values are temperature dependent.

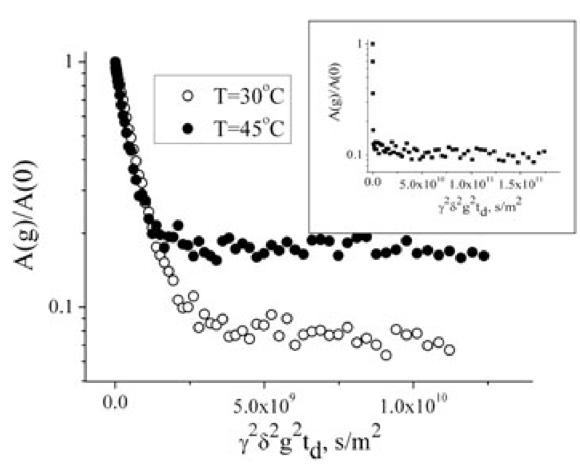

Figure 2. Diffusion decays for HA–D_2O solution (4 wt%) at two temperatures. Diffusion time, 7 ms. The insert shows the diffusion decay for this solution at 45°C, and the abscissa is extended by a factor of 15.

Table 2. Self-diffusion coefficients of water, HA and BSA molecules in HA–D_2O, BSA–D_2O and HA–BSA–D_2O systems

System studied		30 °C	45 °C
4 wt% of HA in D_2O	D_{HA}, m^2/s	$< 10^{-13}$	$\sim 10^{-13}$
	D_{water}, m^2/s	$(1.55 \pm 0.014)\cdot10^{-9}$	$(2.3 \pm 0.05)\cdot10^{-9}$
4 wt% of BSA in D_2O	D_{BSA}, m^2/s	$(4.8 \pm 0.12)\cdot10^{-11}$	$(1.1 \pm 0.02)\cdot10^{-10}$
	D_{water}, m^2/s	$(1.8 \pm 0.05)\cdot10^{-9}$	$(2.6 \pm 0.1)\cdot10^{-9}$
4 wt% of BSA and 0.25% of HA in D_2O	D_{BSA}, m^2/s	$(3.1 \pm 0.19)\ 10^{-11}$	$(5 \pm 0.06)\cdot10^{-11}$
	D_{water}, m^2/s	$(1.38 \pm 0.3)\ 10^{-9}$	$(2.5 \pm 0.17)\ 10^{-9}$
4 wt% of BSA and 0.75% of HA in D_2O	D_{BSA}, m^2/s	—	$(3.2 \pm 0.07)\cdot10^{-11}$
	D_{water}, m^2/s	$(1.7 \pm 0.27)\ 10^{-9}$	$(2.1 \pm 0.17)\ 10^{-9}$

The self-diffusion of macromolecules in solutions is influenced by a number of factors, such as macromolecular collisions, [13] self-association [14] and network formation [15], that may lead to decreasing diffusion coefficients in a time-dependent manner. Each of these processes could take place in the HA-water system. For this reason, diffusion was measured at three diffusion times: 7, 20 and 33 ms. At longer diffusion times, measurements were problematic because the signal from the HA protons decayed completely because of T_1 relaxation. As was observed in Fig. 2, diffusion decays were of the two-component type. The initial part, caused by the appearance of water, does not change its slope with diffusion time and shows diffusion coefficients in the range of $2.3\cdot10^{-9}$–$2.43\cdot10^{-9}$ m^2/s at 45 °C. The slow-decaying part is caused by the decrease in the apparent population of HA because the T_1 of HA is shorter than that of water. However, the slope of this part does not change significantly, remaining $\leq 10^{-13}$ m^2/s. A time dependence of SDC of HA was not observed. This may be caused by instrumental limitations, for example, the relatively small range of diffusion times and the large error when estimating the mean SDC for HA. Thus, it is not possible to state that there is direct proof of gel formation in the system HA–water, such as restriction of diffusion of HA molecules. However, the diffusion coefficients obtained for HA are typical for macromolecules in gels, $< 10^{-12}$ m^2/s. [16]

Diffusion in a BSA–D_2O system

The second component of the system of interest is BSA. Typical diffusion decays for the BSA–D_2O solution are shown in Fig. 3. These decays also demonstrate two-component forms, but the slopes of the slower-decaying parts corresponding to BSA are a much steeper incline compared with those for HA (Fig. 2). The results of the two-component analysis are shown in Table 2. The diffusion of water is similar to that in HA–water, but the SDC of BSA is two orders higher than that of HA. No diffusion time dependence of the SDC of BSA was observed. The apparent populations of the DD components corresponding to BSA are around 0.9. This is much higher than that in HA–water because BSA is a globular protein, with a developed rotational mobility and a slower T_2 NMR relaxation of protons.

The diffusion of BSA in aqueous solutions has been reported by several authorities, for example. [12,17–19] The concentrations of BSA studied here, 1, 2 and 4 wt%, correspond to volumes concentrations of 0.007, 0.015 and 0.03, respectively. According to the concentration dependence of self-diffusion coefficients of BSA, all three of these concentrations, appeared to show the diffusion characteristics of a dilute protein solution. [12] In this regime, the effect of intermolecular collisions is negligible, and the protein molecule can be described as similar to a hard Brownian non-interacting particle. At the studied pH, the peptide is present in monomeric form. [20] The SDC of a protein in the dilute solution does not depend on the protein

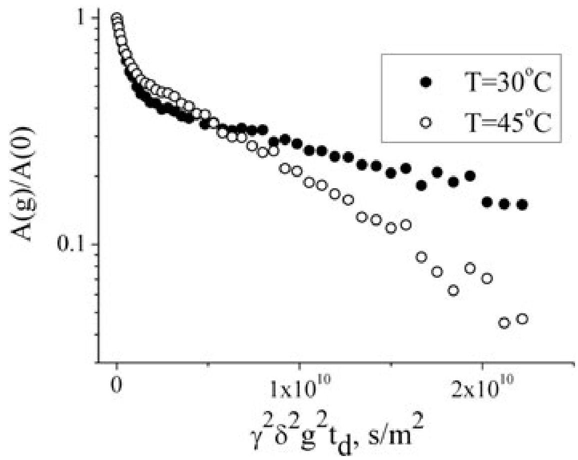

Figure 3. Diffusion decays for bovine serum albumin (BSA)–D_2O solution (4 wt%) at two temperatures. Diffusion time, 7 ms.

concentration as was observed in our measurements with BSA. The values of D for BSA in our study, obtained at 30 °C, coincide with those obtained by Nesmelova *et al.* [12]

Diffusion in a BSA–HA–D_2O system

Figure 4 shows diffusion decays in the system with 4% BSA and different concentrations of HA at 45 °C. These are typical decays for the systems studied at various compositions as presented in Table 1. Again, the decays are of the two-component type with no diffusion-time dependence. Because of the faster T_2 relaxation of HA protons, the slower part of the decay, which is associated with

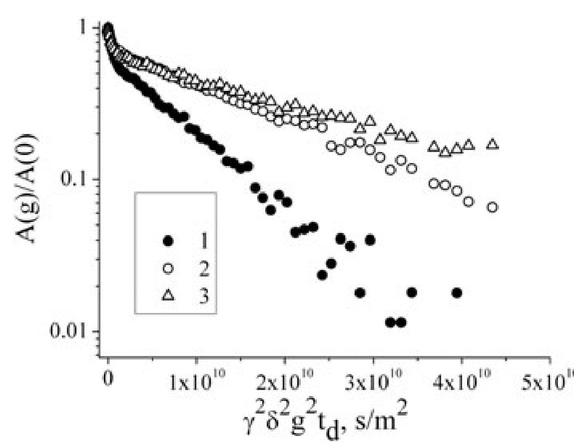

Figure 4. Diffusion decays in BSA–HA–D_2O systems: (1) 4 wt% BSA; (2) 4 wt% BSA and 0.25% HA; 4 wt% BSA and 0.75% HA. Temperature, 45 °C.

the polymer, is mainly caused by BSA diffusion. This part gradually decays more slowly as the concentration of HA increases. Thus, the SDC of BSA decreases. Similar dependences were observed for different BSA concentrations at temperatures of 30 °C and 45 °C. The results of the analysis of the diffusion decays at 45 °C are shown in Fig. 5 and Table 2. The figure demonstrates that the BSA diffusion in the BSA–HA–D_2O system decreases as the concentration of HA increases from 0 to 0.75 wt%.

Discussion

Our experimental data unambiguously demonstrate that HA decreases the translational mobility of BSA molecules. The following factors that may produce this effect are suggested:

1. HA binds to water, thereby decreasing the translational mobility of water in the studied system.
2. HA molecules form spatial, immobile barriers for translational diffusion of BSA molecules.
3. HA and BSA molecules form a complex, which leads to the lower diffusion of BSA molecules compared with a BSA aqueous solution.

Let us discuss these three hypotheses in more detail.

1. Effect of viscosity changes in the presence of HA. HA can bind large quantities of water, which increases its weight by three orders in magnitude. [21] It is possible then that the translational mobility of the water in the system is decreased because of its interaction with HA, and thus, the water viscosity increases. The effect of water viscosity on protein diffusion can be estimated using the known Stokes–Einstein equation

$$D = \frac{kT}{6\pi\eta R_H}, \tag{2}$$

where k is the Boltzmann constant, T is the temperature, η is the viscosity, and R_H is the hydrodynamic radius of the protein molecule. This type of equation is usually applied to the diffusion of globular molecules such as lysozyme and BSA. [12,22] Viscosity is a macroscopic parameter that reflects changes in translational mobility occurring at the molecular level. Nevertheless, the viscosity of the HA solution is high because of the presence of the polymer network, whereas the 'local viscosity' (or 'microviscosity') of the water in the system, which influences the diffusion of the BSA molecules, could be much lower. A parameter more directly related to the 'microviscosity' of water than the bulk viscosity is the water diffusion coefficient in the system. So to estimate the change of 'microviscosity' of water in the presence of HA, the change in the SDC of the water should be estimated. This is best achieved by comparing the SDCs of water in HA solutions and in BSA solutions (Table 2). Surprisingly, the decrease in the SDC of water in a solution of BSA and in a gel-forming system of HA–water is of approximately the same order. For protein solutions, this is typical and is caused by the exchange of less hindered water (not bound to macromolecules) and more hindered water (bound). [23] One of the probable reasons for the decrease of the SDC of albumin (Fig. 5) could be the decrease of the SDC of water. Let us discuss this possibility.

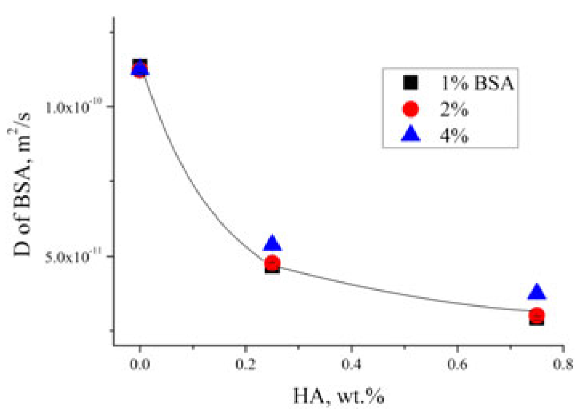

Figure 5. Mean BSA diffusion coefficients in BSA–HA–D_2O systems at three concentrations of HA and three concentrations of BSA. Temperature, 45 °C. The connecting line was drawn by hand.

The 'microviscosity' of water changes inversely to the SDC of water, as it follows from Eqn (2). Therefore, under conditions where other parameters remain unchanged, when the SDC of water is decreased, by 3% for example, the viscosity will change proportionally, and the SDC of BSA will change by a similar value. However, as seen from Fig. 5, the SDC of BSA decreases by a factor of 1.5–2.2. Such a big change cannot be explained purely by the change in water 'microviscosity' in the presence of HA.

2. Effect of HA barriers on the diffusion of BSA molecules. Let us hypothesize that the presence of HA at such a low concentration, up to 0.75 wt%, has no effect compared with that of the BSA concentration on the SDC of BSA. That is, the decrease in the SDC of BSA is only caused by collisions of BSA molecules with the network of HA molecules.

An initial contradiction to this hypothesis was encountered when comparing signal amplitudes from macromolecules in BSA–D_2O and BSA–HA–D_2O (0.75% HA) systems. For the latter system, the decrease in amplitude was almost threefold, which cannot be explained by collisions alone, without the involvement of other types of interaction.

In addition, the effect of spatial restrictions on the SDC of solute molecules has been analyzed in a number of works, for example. [24,25] It has been determined that the volume fraction of different-shaped obstacles determines the decrease in the SDC of the solute. [24] The HA monomer has a molecular mass of $m_U = 379$ g/mol, and based on the geometry of D-glucuronic acid and D-N-acetylglucosamine monomers,[2] it is estimated to be 14.5 Å long and around 6 Å wide. Therefore, the volume of the HA monomer is $v_U \sim 395$ Å.[3] The volume fraction of HA can be estimated as

$$\phi_{HA} = \frac{m_{HA} v_U}{m_U} \cdot \left(\frac{m_w}{\rho_w} + \frac{m_{HA} v_{HA}}{m_U} \right)^{-1}, \tag{3}$$

where m_{HA} and m_w are the mass of HA and the mass of water in the sample, respectively, and ρ_w is the water density. For small mass fractions of HA (ω_{HA}), the equation can be written as

$$\varphi_{HA} \approx \omega_{HA} \frac{v_U \rho_w}{m_U}. \tag{4}$$

The volume fractions occupied by HA molecules at concentrations of 0.25% and 0.75% were calculated, producing volume fractions of 0.0076 and 0.0228, respectively. In papers [24,25] describing the effects of immobile obstacles of spherical, cylindrical and circular shape, the decrease in the SDC for volume fractions of obstacles similar to that of HA in our systems was less 5%. Our estimations also show that decreases in the SDC of BSA by factors of two and three (our case) require around 0.2 and 0.25 volume fractions of obstacles, respectively. In cases of mobile obstacles (similar to our HA with a SDC of ~10^{-13} m^2/s, Table 2), the effect of obstacles on diffusion is significantly lower.

Therefore, it can be concluded that the two factors discussed above have only small effects on the decrease in the SDC of BSA in the presence of HA.

3. Complex formation of BSA with HA. The ability of BSA to complex with HA has been established previously;[6–8,26] however, the thermodynamic nature of the complexation of peptides with polysaccharides is still a matter of debate. A number of authors report that the electrostatic interaction is enthalpically driven because of a reduction of the electrostatic energy of the system. [27,28] Other authors argue that the complexation is entropically driven because of the release of counterions and water molecules. [29] In the electrostatic interaction, the charge of HA is negative (one negative charge per monomer), and at pH $<$ 4, the charge of BSA is positive. [30] However, at physiological pH, [30] the charge of the BSA molecule as a whole is negative, and the HA and BSA continue to interact with each other. This could be caused by the charge distribution on the surface of BSA molecule, which has positive charges in some places.[31]

Figure 5 shows that the SDC decreases by a factor of 2.2 when the concentration of HA varies from 0 to 0.25% for all BSA concentrations when the weight ratio of BSA/HA was varied from 1.33 to 16. At the next concentration interval of 0.25–0.75% HA, the SDC of BSA decreases by a factor of 1.5 at all concentrations of BSA. It is known that HA favors an elongated configuration under the studied conditions because of the electrostatic repulsion between the carboxylate groups,[2] whereas the BSA molecule is globular. Therefore, several protein molecules may attach to one polysaccharide, and, as it follows from the data, the presence of HA and the density of the HA network are more effective for BSA diffusion than the BSA concentration.

The data obtained provide indirect proof of formation of a BSA–HA complex, with the detail of the mechanisms of this process and the structure of the complex being beyond the scope of the technique used here. It is evident that the mechanism is not simple. However, PFG NMR leads to some conclusions about the dynamic character of the complex and its parameters. As the SDC of BSA in the complex is still much higher than that of HA, and as separate signals from 'free' BSA molecules and BSA molecules interacting with HA in the diffusion decays were not observed (Fig. 4), it can be stated that exchange between the 'free' and 'bound' states of BSA molecules occurs much faster than the minimum diffusion time of the PFG NMR experiment, 7 ms. This corresponds to the 'fast exchange' conditions in terms of NMR. Let the lifetime of a 'bound' state be τ_b, and the lifetime in the 'free' state be τ_f. $\tau_b,\tau_f << t_d$. Because of the SDC of BSA being higher than that of HA, the SDC of the complex, D_b, is close to the SDC of HA. The fraction of BSA molecules in the complex (which also means a fraction of time the BSA molecules reside in the complex) is

$$P_b = \frac{\tau_b}{\tau_b + \tau_f}, \tag{5}$$

and the fraction of 'free' BSA molecules is

$$P_f = 1 - P_b. \tag{6}$$

The apparent (measured) SDC of BSA molecules is

$$D_{\exp} = P_f \cdot D_f + P_b \cdot D_b. \tag{7}$$

Thus,

$$P_b = \frac{D_{\exp} - D_f}{D_b - D_f}. \tag{8}$$

Now, the fraction of 'bound' BSA in the BSA–HA complex can be estimated. The SDCs of molecules in BSA–HA–D_2O system were obtained from slopes of slowly decaying parts of diffusion decays (Fig. 4 and similar decays for 30 °C).

At 30°C, the SDC of BSA in solution is $4.8 \cdot 10^{-11}$ m^2/s, the SDC of HA in solution is ~ $1 \cdot 10^{-13}$ m^2/s and the SDC of BSA in the BSA–HA–D_2O system at 0.25% HA is $3.1 \cdot 10^{-11}$ m^2/s. Consequently, $P_b = 0.36$.

At 45°C, the SDC of BSA in solution is $1.1 \cdot 10^{-10}$ m^2/s, the SDC of HA is ~$1 \cdot 10^{-13}$ m^2/s and the SDC of BSA in the BSA–HA–D_2O system at 0.25% HA is $5 \cdot 10^{-11}$ m^2/s. Consequently, $P_b = 0.55$.

At 45°C, the SDC of BSA in solution is $1.1 \cdot 10^{-10}$ m^2/s, the SDC of HA is $1 \cdot 10^{-13}$ m^2/s, and the SDC of BSA in the BSA–HA–D_2O system with 0.75% HA is $3.2 \cdot 10^{-11}$ m^2/s. Consequently, $P_b = 0.7$.

Thus, the experimental results can be accurately described, keeping in mind the complex formation of BSA with HA. Using the PFG NMR technique, it is possible to estimate the range of the lifetimes and fractions of BSA molecules in the formed complexes.

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